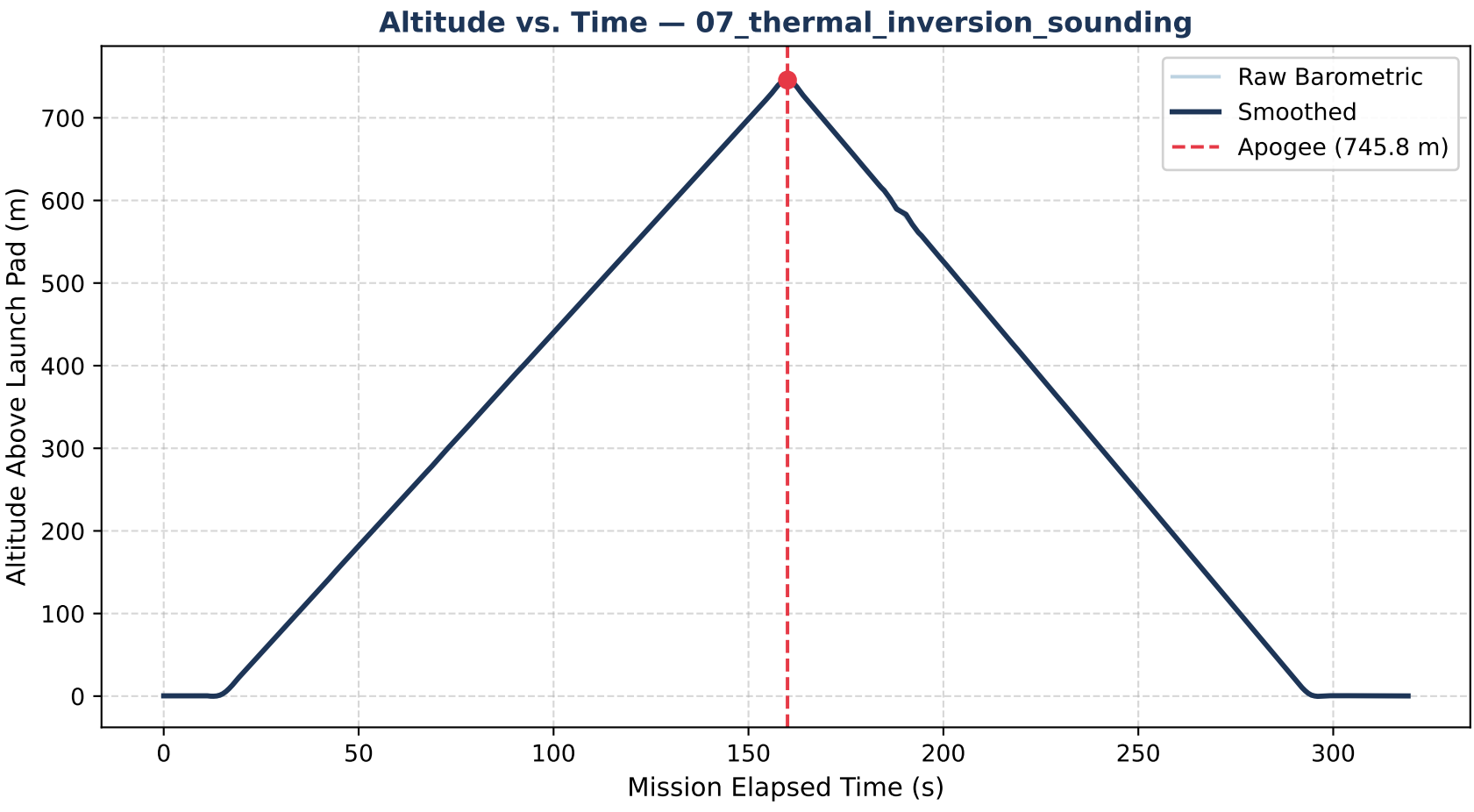
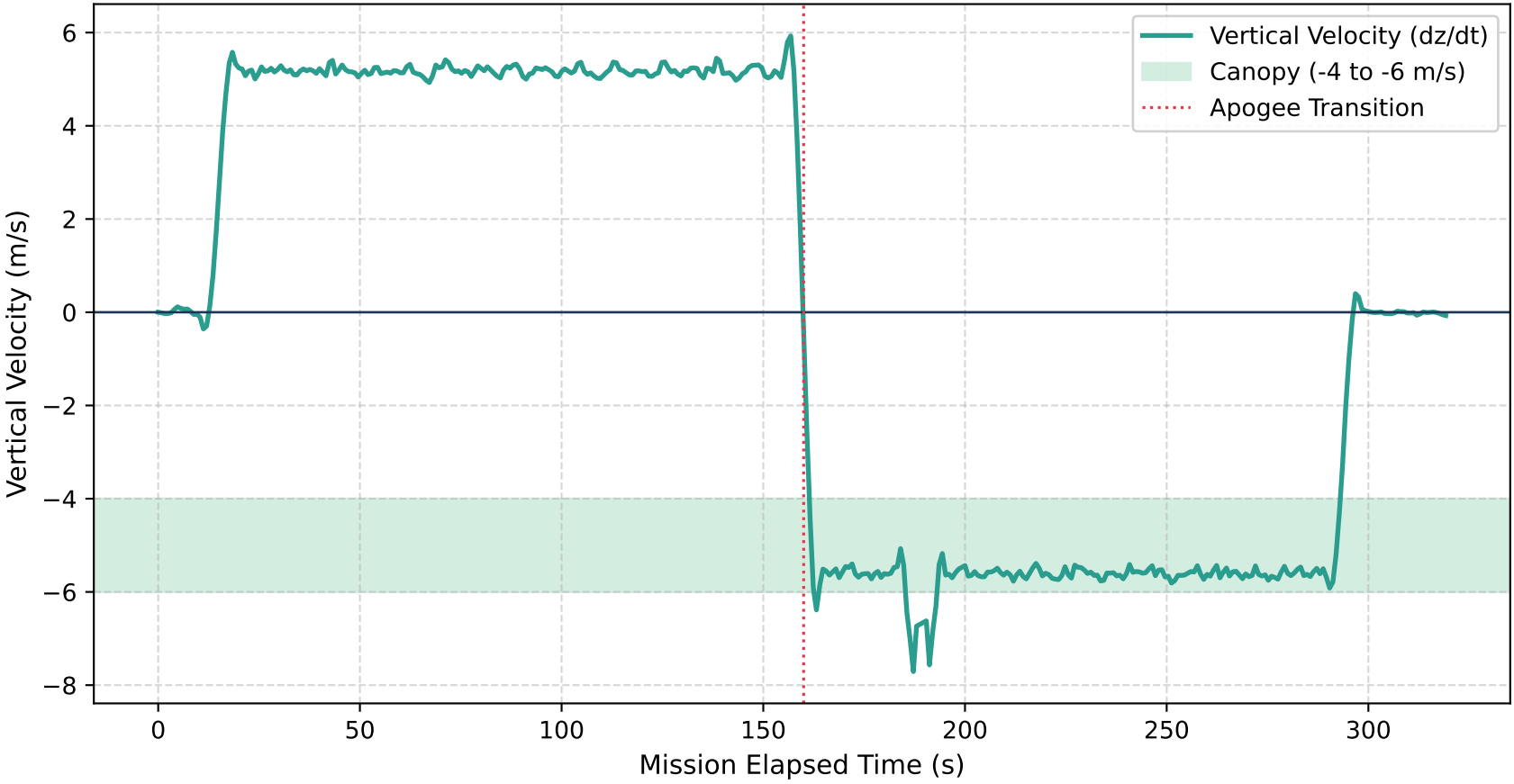
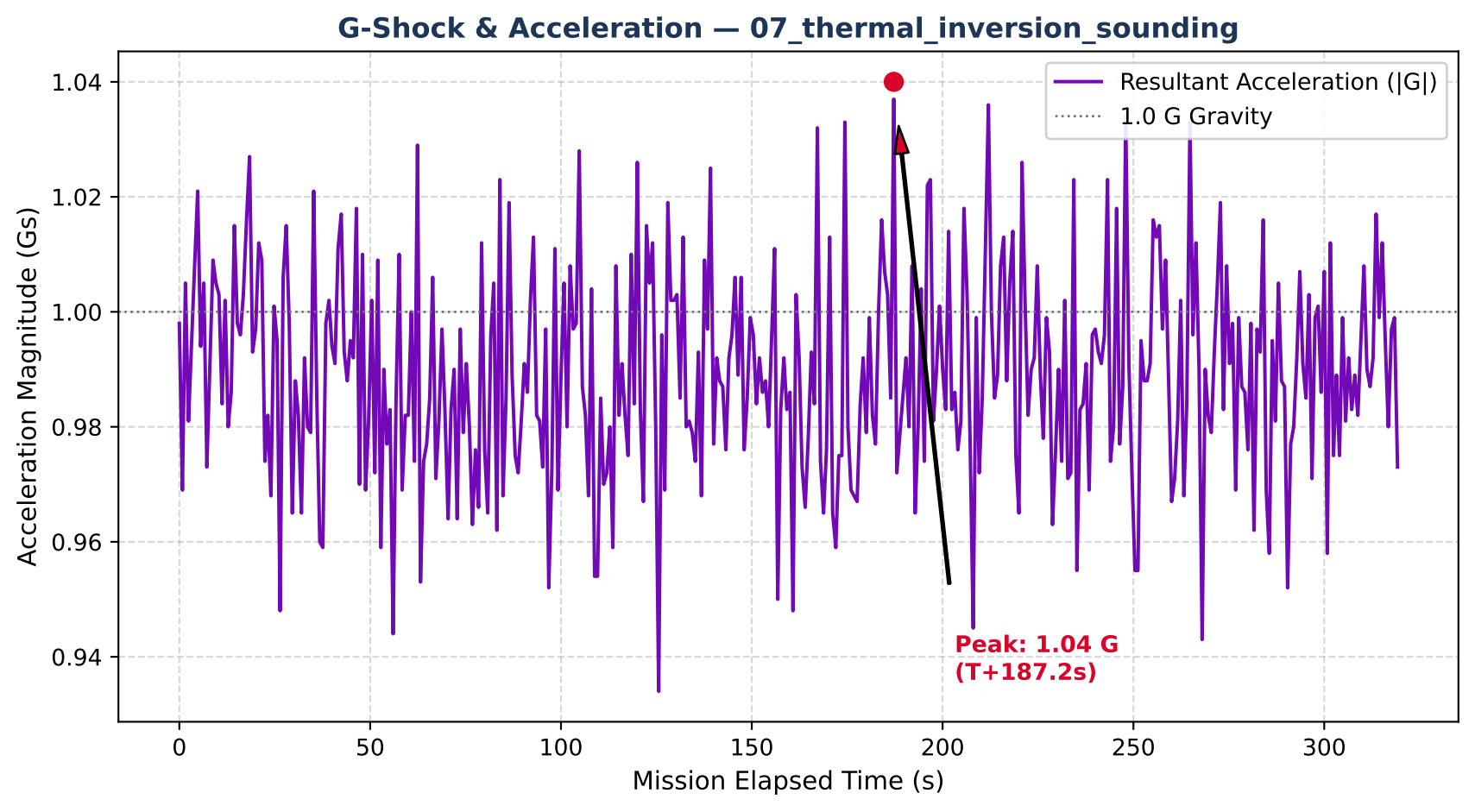
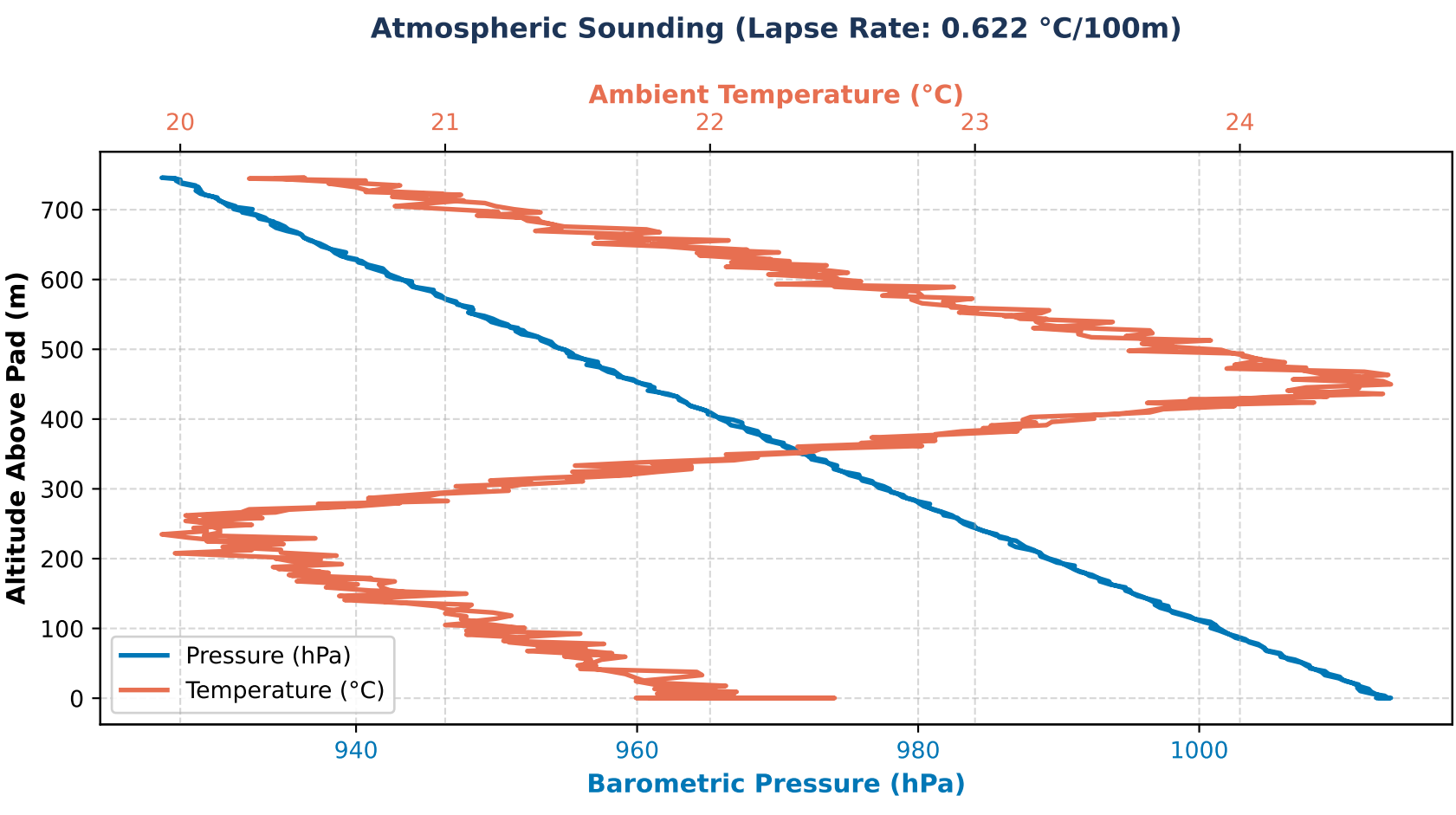
Cognitive CanSat - Mission Summary: 07_thermal_inversion_sounding

Flight Dynamics Parameter	Calibrated Value
Mission Name	07_thermal_inversion_sounding
Apogee (m)	745.76
Time to Apogee (s)	160.01
Max Ascent Velocity (m/s)	5.93
Max Descent Velocity (m/s)	-7.71
Max G-Shock (Gs)	1.04
Touchdown Shock (Gs)	1.02
Mission Duration (s)	319.2
Min Temperature (°C)	19.93
Max Temperature (°C)	24.57
Min Pressure (hPa)	926.17
Max Pressure (hPa)	1013.64
Lapse Rate (°C/100m)	0.622
Telemetry Frames	398
Landing Lat	22.57574
Landing Lon	88.369098


Altitude vs. Time — 07_thermal_inversion_sounding
Raw Barometric
Smoothed
Apogee (745.8 m)
Altitude Above Launch Pad (m)
0
100
200
300
400
500
600
700
0
50
100
150
200
250
300
Mission Elapsed Time (s)


Vertical Velocity Profile — 07_thermal_inversion_sounding
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
6
4
2
0
−2
−4
−6
−8
0
50
100
150
200
250
300
Mission Elapsed Time (s)


G-Shock & Acceleration — 07_thermal_inversion_sounding
Resultant Acceleration (|G|)
1.0 G Gravity
Peak: 1.04 G
(T+187.2s)
Acceleration Magnitude (Gs)
Mission Elapsed Time (s)
1.04
1.02
1.00
0.98
0.96
0.94
0
50
100
150
200
250
300


Atmospheric Sounding (Lapse Rate: 0.622 °C/100m)
Ambient Temperature (°C)
20
21
22
23
24
Altitude Above Pad (m)
700
600
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
940
960
980
1000
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 07_thermal_inversion_sounding

